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Oefeningenreeks 2D nr. 10.12

$$\begin{aligned} & \lim_{x \rightarrow -\frac{1}{2}} \frac{-6x^2 - x + 1}{-10x^2 - x + 2} \\ & -6x^2 - x + 1 = -6 \left(x + \frac{1}{2} \right) \left(x - \frac{1}{3} \right) \\ & -10x^2 - x + 2 = -10 \left(x + \frac{1}{2} \right) \left(x - \frac{2}{5} \right) \\ & \lim_{x \rightarrow -\frac{1}{2}} \frac{-6x^2 - x + 1}{-10x^2 - x + 2} = \lim_{x \rightarrow -\frac{1}{2}} \frac{-6 \left(x + \frac{1}{2} \right) \left(x - \frac{1}{3} \right)}{-10 \left(x + \frac{1}{2} \right) \left(x - \frac{2}{5} \right)} \\ & = \lim_{x \rightarrow -\frac{1}{2}} \frac{-3x + 1}{-5x + 2} = \frac{5}{9} \end{aligned}$$

References